

[BUG] vqe_solver with default UCCSD Ansatz inserts two copies of entang

<https://github.com/goodchemistryco/Tangelo/issues/125>

ROW 6

[BUG] vqe_solver with default UCCSD Ansatz inserts two copies of entangler? #125

✓ Closed



elloyd-1qbit opened on Feb 12, 2022

...

Issue: Bug Report

Current Behavior

The H2 molecule in sto-3g basis with scBK qubit mapping should be solved with a single entangler with the UCCSD ansatz. Currently the algorithm seems to insert two copies of the same entangler.

```
H2 = [('H', (0, 0, 0)), ('H', (0, 0, 0.74137727))]
mol_H2 = SecondQuantizedMolecule(H2, q=0, spin=0, basis="sto-3g")

from tangelo.algorithms.variational import VQESolver

vqe_options = {"molecule": mol_H2, "qubit_mapping": "scbk", "up_then_down": True}
vqe_solver = VQESolver(vqe_options)
vqe_solver.build()
vqe_solver.simulate()
print(vqe_solver.optimal_circuit)
```

returns

```
X      target : [0]
X      target : [1]
RX     target : [0]   parameter : 1.5707963267948966
RZ     target : [0]   parameter : 12.566261789411744 (variational)
RX     target : [0]   parameter : 10.995574287564276
RX     target : [1]   parameter : 1.5707963267948966
RZ     target : [1]   parameter : 12.566261789411744 (variational)
RX     target : [1]   parameter : 10.995574287564276
RX     target : [0]   parameter : 1.5707963267948966
H      target : [1]
CNOT   target : [1]   control : [0]
RZ     target : [1]   parameter : 0.11304437992941474 (variational)
CNOT   target : [1]   control : [0]
H      target : [1]
RX     target : [0]   parameter : 10.995574287564276
H      target : [0]
RX     target : [1]   parameter : 1.5707963267948966
CNOT   target : [1]   control : [0]
RZ     target : [1]   parameter : 0.11304437992941474 (variational)
CNOT   target : [1]   control : [0]
RX     target : [1]   parameter : 10.995574287564276
H      target : [0]
```

This circuit does return the right energy.

```
from tangelo.linq import Simulator
sim=Simulator()
print(sim.get_expectation_value(vqe_solver.qubit_hamiltonian, vqe_solver.optimal_circuit))
```

returns -1.1372704157729252

Was this happening before? I can look into the `BuiltInAnsatz.UCCSD` or potentially the changes in the `Circuit` class



elloyd-1qbit added **bug** on Feb 12, 2022

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```
print(sim.get_expectation_value(vqe_solver.qubit_hamiltonian, vqe_solver.optimal_circuit))
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elloyd-1qbit added **bug** on Feb 12, 2022

elloyd-1qbit assigned [ValentinS4t1qbit](#), [alexfleury-sb](#), [JamesB-1qbit](#) and [elloyd-1qbit](#) on Feb 12, 2022



alexfleury-sb on Feb 14, 2022 · edited by alexfleury-sb

Edits ▾ Collaborator ...

I am getting `-1.1372704` with both `CCSDSolver` and `FCISolver`, so I am not sure about the energy comment.

For the circuit, it all comes from `openfermion.circuits.uccsd_singlet_generator` in `UCCSD._get_singlet_qubit_operator` method. We are not removing redundant terms (different phases).

For H2, the singlet fermionic operator results in

```
5.4412555926570146e-05 [0^ 2] +  
-0.0565221899617258 [0^ 2 1^ 3] +  
5.4412555926570146e-05 [1^ 3] +  
-0.0565221899617258 [1^ 3 0^ 2] +  
-5.4412555926570146e-05 [2^ 0] +  
0.0565221899617258 [2^ 0 3^ 1] +  
-5.4412555926570146e-05 [3^ 1] +  
0.0565221899617258 [3^ 1 2^ 0]
```

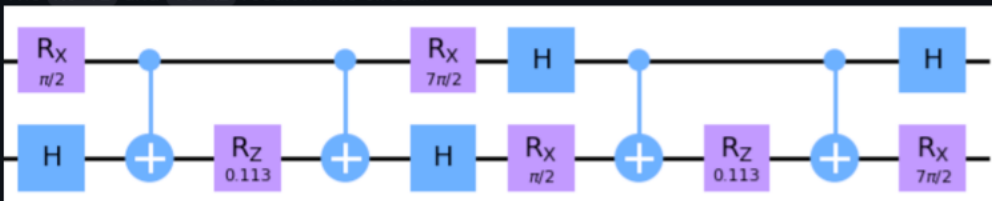


And get converted to (with `scBK`):

```
0.0565221899617258j [X0 Y1] +  
(-0-5.4412555926570146e-05j) [Y0] +  
0.0565221899617258j [Y0 X1] +  
(-0-5.4412555926570146e-05j) [Y1]
```



The `X0 Y1` and `Y0 X1` result in the circuit:



In this case, it is redundant. This is one the reason the `UCC1` and `UCC3` exist, as they are removing those terms. There are many things we could do with symmetries for resource reduction (we already have some code ready-ish).



elloyd-1qbit on Feb 15, 2022

Contributor Author ...

Ah gotcha, didn't take a close enough look at the circuit, thanks!



elloyd-1qbit closed this as **completed** on Feb 15, 2022



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